The News Article Classification Problem

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***Abstract*—In this report, we propose a method to predict the category of news articles leveraging their content, sources, and PageRank. We demonstrate that our approach, based on Linear Support Vector Machines, is robust and achieves competitive results. Finally, we provide a deep analysis of the model's performance and learned characteristics.**

I. Problem Overview

The proposed competition centers on the classification of news articles into seven distinct categories: International News, Business, Technology, Entertainment, Sports, General News, and Health. The dataset is divided into two parts:

- a *development set*, containing $79,996$ labeled articles;
- an *evaluation set*, comprised of $20,000$ unlabeled articles.

Each row includes the article's source, title, and content, alongside its associated PageRank and publication date. Based on the development set, we observe that the dataset is significantly unbalanced. While the intermediate classes contain roughly $11,000$ samples each, "International News" is heavily over-represented ($23,542$ articles), whereas "Health" is under-represented with only $3,102$ instances.

It is possible to notice some interesting patterns in the sources of the articles. In fact, looking at Figure 7 we notice discrepancies in source distributions across categories. For instance, while Yahoo is the predominant publisher for International News, it plays a marginal role in General News. Additionaly, Rednova appears to be a major player only within the Technology category.

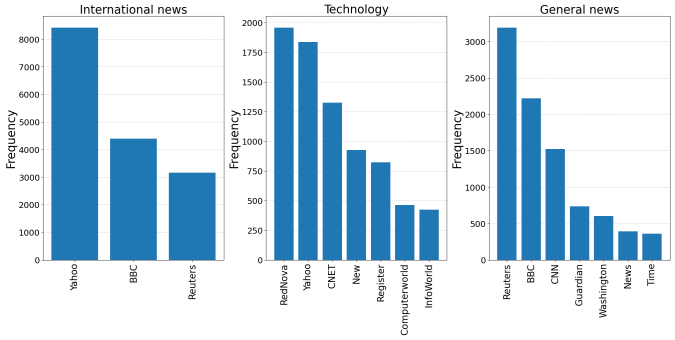


Figure 1. Source distribution across different categories

Furthermore, some articles contain auxiliary information, such as image descriptions or hyperlinks to external sources. Links appear to provide valuable insights, often explicitly mentioning the geographic region concerned or specific keywords related to the topic of the article (e.g., "movie", "baseball", etc.).
To get a better understanding of the dataset content, we can examine the most frequent words in the articles and titles in Table 1, excluding metadata and stopwords.

Table I
Most frequent words in article bodies and titles

Category		Most frequent words
International news	*Body*	said, reuters, **president**, **afp**, new
	Title	**afp**, reuters, **iraq**, **bush**, says
Business	*Body*	reuters, said, new, **york**, **company**
	Title	reuters, **stocks**, **oil**, **profit**, new
Technology	*Body*	said, new, **company**, **microsoft**, **software**
	Title	new, **microsoft**, reuters, **google**, **web**
Entertainment	*Body*	reuters, new, said, two, **film**
	Title	reuters, new, says, iraq, **film**
Sports	*Body*	new, first, **night**, **team**, **victory**
	Title	**win**, **game**, **wins**, **cup**, back
General news	*Body*	reuters, said, new, **president**, **iraq**
	Title	**iraq**, **bush**, new, says, **iran**
Health	*Body*	reuters, new, **health**, said, **cancer**
	Title	**cancer**, **flu**, **study**, **drug**, **health**

Some words seem to be really common across all categories, like "reuters" or "new", and others that are specific to each category, such as "software" or "cancer". These specific words make the classes distinguishable from one another. This is particularly evident for the category "Health", while it seems less clear for the categories "International News", "Entertainment", and "General News", which appear more difficult to distinguish. Therefore, this table suggest that a tf-idf representation of the document is particularly appropriate for solving this kind of problem.

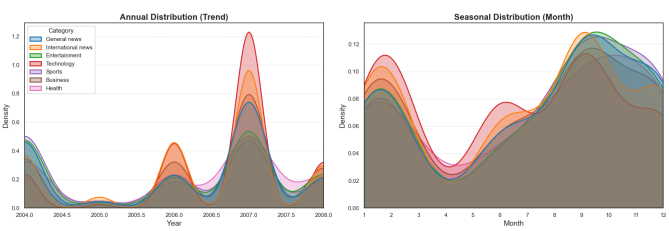


Figure 2. Distribution of articles pubblication across months and years

To conclude this overview, we observe that the distributions of the publication months and years (Figure 2) are quite similar across categories, suggesting they are not particularly relevant. On the other hand, the PageRank feature seems useful to recognize the class "Technology", which is the only one to have a PageRank ≤ 3 as shown in figure 3.



Figure 3. Distribution of the Page Rank distribution among multiple classes

II. Proposed Approach

A. Preprocessing

First, the dataset is balanced to ensure robust model training. Specifically, we perform downsampling on the "International News" class, reducing it by approximately 7,000 examples. This approach is adopted to mitigate potential bias toward this category, particularly as the "General News" class exhibits highly similar characteristics. On the other hand, no oversampling is applied to the "Health" class. We assume that the model possesses a sufficient number of examples to achieve accurate classification, given that this category is highly distinct and easily recognizable, as demonstrated in the initial data overview.

Then, we focus on the preprocessing of the textual data. Each title and article is processed through a pipeline involving lowercasing, removal of punctuation and stop-words, lemmatization, and tokenization. We extract relevant metadata using BeautifulSoup [1], a Python library for HTML processing, which is particularly well suited for our objectives. This information is then appended to the original article corpus. To represent the articles and titles as feature sets, we employ a Bag-of-Words representation combined with a TF-IDF weighting scheme. For a given term t in a document d belonging to a set of documents D, the TF-IDF value is calculated as follows:

$$\text{tf-idf}(t, d, D) = f_{t,d} \cdot \log\left(\frac{|D|}{|\{d \in D : t \in d\}|}\right) \quad (1)$$

where $f_{t,d}$ is the frequency of term t in document d. Under this scheme, words that are highly representative of a specific document (e.g., "cancer" or "baseball") receive a higher weight compared to common words (e.g., "say" or "reuters"). To capture relevant word patterns, such as "New York", we also encode sequences of tokens of length ℓ in a similar fashion. The value of ℓ is left as an hyperparameter to be tuned. Furthermore, we encode titles and articles using separate schemes; this enables the model to better capture the patterns that are peculiar to each specific section of the text.

As previously seen, the article sources also provide valuable information. Consequently, we encode the sources that published a certain percentage p of the articles using one-hot encoding, grouping all other sources into an "other" category. To choose the value of p, we trained an unoptimized version of the model (described in B)). Figure 4 shows that with $p = 95\%$ we obtain slightly better performances by encoding a relatively small number of sources.

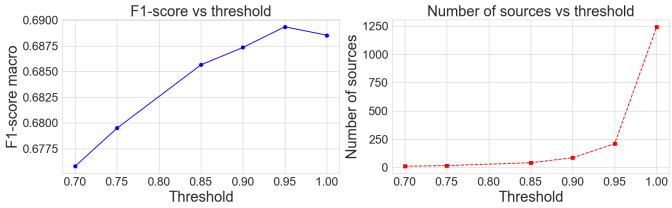


Figure 4. F1-score achieved by the model compared to the number of sources encoded

This approach reduces the number of features used to represent the sources and makes the model more robust to sources that it has not encountered during training.

Finally, we apply standardization to the PageRank feature. This step is crucial because the Linear SVM selected in Section B) is a distance-based model, and unscaled features with larger magnitudes would otherwise dominate the optimization process. Figure 5 summarizes the preprocessing steps we used.

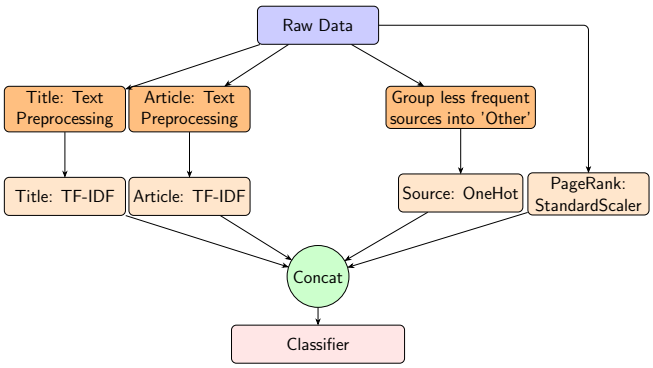


Figure 5. Preprocessing pipeline

B. Model selection

In our initial experiments, we employed Random Forest classifiers. However, due to the high dimensionality and spar-

sity of the feature matrix, the computational cost of training proved to be prohibitive. Consequently, we opted for an alternative approach. As demonstrated by Joachims [2], Support Vector Machines (SVMs) are particularly effective for text classification tasks dealing with such sparse representations. Moreover, while non-linear kernels (e.g., RBF) provide flexibility, they incur a training complexity of approximately $O(N^2)$, which becomes intractable for large datasets. In contrast, Joachims [3] demonstrates that Linear SVMs can be trained in $O(N)$, scaling linearly with the number of samples. Because of this, we decided to use a LinearSVM model.

C. Hyperparameters tuning

Our model selection process is divided into two main stages:

- **TF-IDF Hyperparameter Tuning**: We optimize vectorization hyperparameters, such as the maximum n-gram length ℓ, independently for the title and the article content.
- **Classifier Hyperparameter Tuning**: We tune the specific parameters of the SVM model (e.g., regularization strength).

To simplify the optimization step, we assume these two steps to be orthogonal; consequently, we tune them independently. To select the best parameters we run a random grid search using a 80/20 train/test split, 2 fold cross validation. Table 2 presents the parameter used for the tuning of the preprocessing, while Table 3 the one used for tuning the model:

Table II
GRID SEARCH PARAMETERS FOR THE ARTICLE TF-IDF VECTORIZER

Parameters	Values
max_df	0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9
min_df	1, 2, 3, 4, 5
ngram_range	(1,1), (1,2), (1,3), (1,4), (1,5)
max_features	25000, 50000, 75000, 100000, 150000, 300000, None

Table III
GRID SEARCH PARAMETERS FOR THE LINEAR SVM

Parameters	Values
C	0.01, 0.1, 0.3, 0.5, 0.7, 1, 1.5, 2
class_weight	None, balanced
tol	1e-4, 1e-8, 1e-12
dual	True, False

III. RESULTS

Table 4 shows the best parameters of the model computed using random grid search. We notice that the optimal parameters for the Title Vectorizer differ from those of the Article Vectorizer. The former appears to encode shorter, rare patterns, whereas the latter captures also longer sequences that occur more frequently, as permitted by the higher `max_df` threshold. In addition, the regularization factor $C = 0.1$ used in the support vector machine allows for a softer margin, helping the model to generalize better on unseen data. Using the parameters, the model obtain a macro-f_1 equals to 0.731.

Table IV
BEST HYPERPARAMETERS OF THE LINEARSVC MODEL

Component	**Parameters**
SVM	C: 0.1, class_weight: balanced, dual: true tol:1e-4
Title Vectorizer	ngram: (1,3), min_df: 1 max_df: 0.3, max_feat: 100k
Article Vectorizer	ngram: (1,5), min_df: 1 max_df: 0.8, max_feat: 100k

On the public leaderboard, the model achieves a macro-F_1 score of 0.732, which is very similar to the one obtained on the training set, proving that our model neither overfits nor underfits the data.

IV. DISCUSSION

The confusion matrix in Figure 6 helps us understand where our model commits misclassification errors. The classes that cause difficulty for our model are International News and General News, which are often confused with one another. This was expected, as we have seen in the overview that those two classes share almost the same most frequent words. Interestingly, the model achieves robust performance in the Health category, despite the limited number of training examples. This is likely due to the fact that this class exhibits a highly distinct vocabulary.

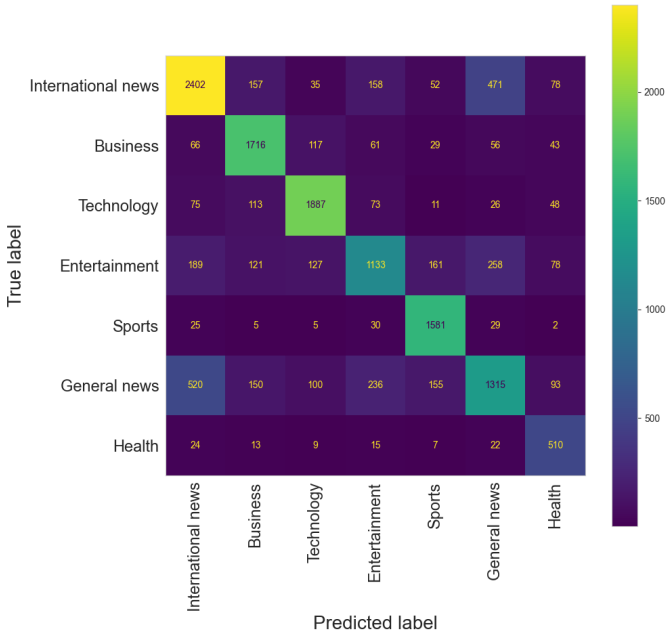


Figure 6. Multiclass confusion matrix

In addition, looking at some misclassified examples, we notice that some of the errors are totally understandable and even a human would have committed them. For example, the General News article "Russia arrests British Council chief: Diplomatic row escalates with arrest of staff including Neil

Kinnock's son" was classified as International News, which is totally reasonable.

To gain insights into the model's decision-making process, we employ an interpretability technique for linear models, as shown in Figure 7. By analyzing the feature weights of the LinearSVM, we can identify the most significant patterns; specifically, features associated with weights of larger magnitude are considered more relevant to the classification task.

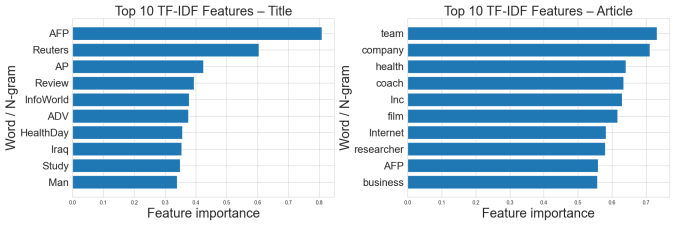


Figure 7. Most important patterns learned by the model

First, we notice that the most important patterns differ between the titles and the articles, proving that our initial assumptions about them showing different features were true. Second, we observe that the most important words learned are highly relevant for classifying the examples for both articles and titles. In the case of the former, we notice that words that represent the names of the categories (e.g., health, business) are learned as relevant; for the latter, the model considers relevant the press agencies that wrote the article, such as AFP (Agence France-Presse) and AP (Associated Press), which are two of the most important international news agencies in the world.

To conclude, this work can be extended in several directions. A primary improvement would involve the adoption of word embedding models to generate features capable of capturing the semantic meaning of titles and articles, overcoming the limitations of the TF-IDF bag-of-words approach. Additionaly, the model's performance could be further enhanced by expanding the hyperparameter search space. Specifically, performing a joint optimization. of both preprocessing and model hyperparameters could uncover more effective combinations that were missed by treating these steps independently.

References

[1] L. Richardson, *Beautiful soup documentation*, Accessed: 2024-05-20, 2023. [Online]. Available: https://www.crummy.com/software/BeautifulSoup/bs4/doc/.

[2] T. Joachims, "Text categorization with support vector machines: Learning with many relevant features," in *European conference on machine learning*, Springer, 1998, pp. 137–142.

[3] T. Joachims, "Training linear svms in linear time," in *Proceedings of the 12th ACM SIGKDD international conference on Knowledge discovery and data mining*, 2006, pp. 217–226.